University of St. Thomas

GCSE 3302 Network Operations and Data Security: Introduction to technical aspects of network operations and issues surrounding the management of data security.

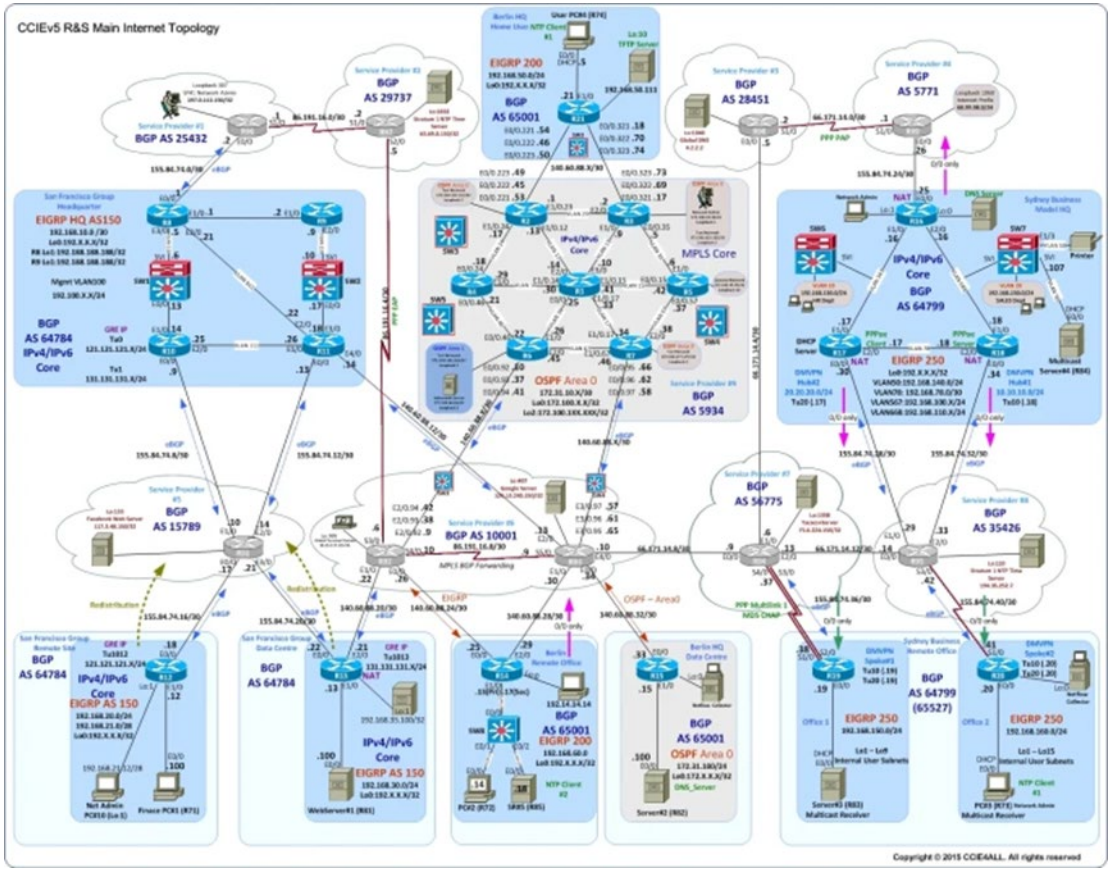
GCSE 3302 GCSE 3302 Network Operations and Data Security: Introduction to technical aspects of network operations and issues surrounding the management of data security.

Session 4 – Networking Equipment:

Routers, switches, servers, ect

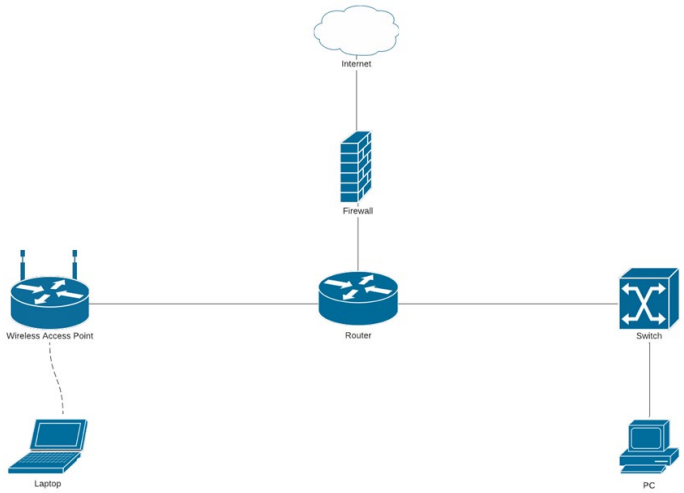
Session Overview

Objective: To be able to identify and understand the purpose and functionality of the different network components.



Network Equipment

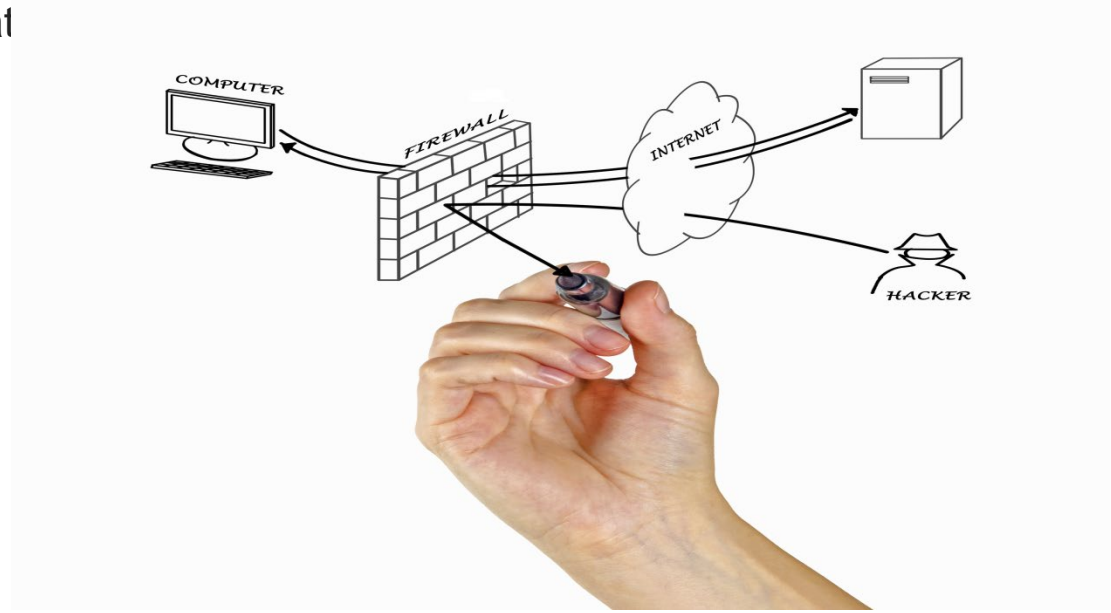
- **Firewalls** safeguard a network against unauthorized access.
- **Routers** are protocol-dependent devices that connect sub-networks or divide a very large network into smaller sub-networks.
- **Switches** connect devices to host computers and allow large numbers of these devices to share a limited number of ports.



Firewalls

Network firewalls protect computer networks against unauthorized use or attack. They permit or deny access to private network devices and applications, and represent an important part of an organization's overall security policy.

Firewalls may be software applications, hardware devices (such as routers), or a combination of both. They include turnkey products that are relatively easy to install as well as complex, multi-layer installations that
administrator.

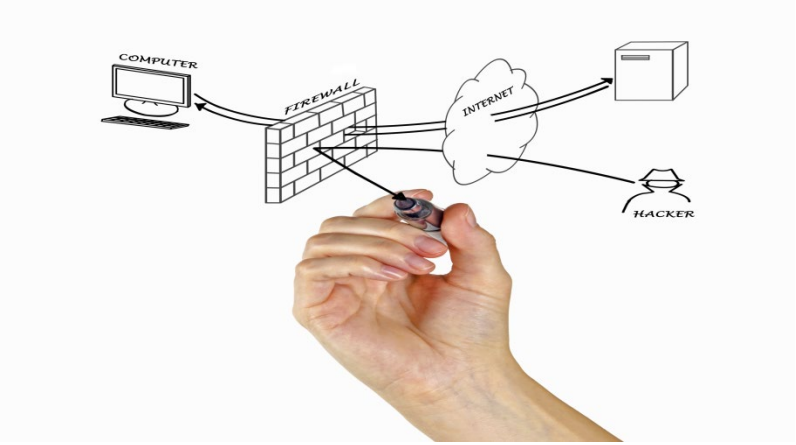


Firewalls

They monitor incoming and outgoing network traffic and decides whether to allow or block traffic based on a defined set of security rules.

Firewalls have been a first line of defense in network security for over 25 years. They establish a barrier between secured and controlled internal networks that can be trusted and untrusted outside networks, such as the Internet.

A firewall can be hardware, software, software-as-a service (SaaS), public cloud, or private cloud (virtual).



Next-generation firewall

A next-generation firewall or third generation firewall technology is built with intelligent, context-aware security capabilities to address advanced security threats at the application level. New security threats are constantly evolving.

NGFW combines standard firewall features like packet filtering and ***stateful inspection*** with other features to provide additional security capabilities.

A next-generation firewall has the capacity to analyze packet data and filter packets based on applications (rather than just their IP headers). In the OSI model, it operates up to layer 7, the application layer, as opposed to earlier firewall technology, which only supported levels 4 through 6. (the transport layer). ***This is a crucial capacity since attacks on the OSI model's levels 4–7 are become more frequent.***

Stateful Firewall:

A stateful firewall examines the contents of data packets as well as the data's properties and communication channels.

They analyze the behavior of data packets and can filter out the suspect data if anything seems out of the ordinary.

A stateful firewall can also monitor the activity of the data and record any patterns of behavior. Even if the suspicious behavior was not manually entered by an administrator, the firewall can detect it and stop the threat if a data packet analysis reveals it.

An internal segmentation firewall (ISFW), which guards particular network segments in the event that malicious code gets inside, is an example of an internal stateful firewall that can be used at the network's edge or internally.

Stateless Firewall

Stateless firewalls employ the source, destination, and other information in a data packet to determine if the data poses a threat.

Administrators must enter these parameters in accordance with guidelines they have previously established. The stateless firewall protocol will identify threats and then restrict or block the data carrying them whenever a data packet deviates from what is deemed acceptable.

What is a router?

A router is a device that connects packet-switched networks or subnetworks. By sending data packets to their intended IP addresses, it manages traffic between different networks and permits several devices to share an Internet connection.

Although there are many different kinds of routers, the majority of them transfer data between LANs (local area networks) and WANs (wide area networks). A LAN is a collection of linked devices confined to a certain region. Typically, a LAN needs just one router.

Network Routers

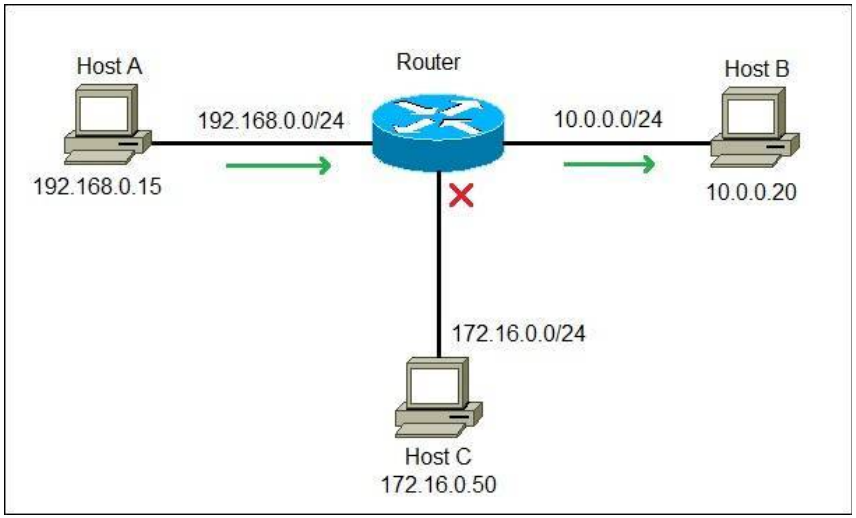
Network routers are protocol-dependent devices that connect subnetworks, or that break down a large network into smaller subnetworks.

Routers connect multiple networks together. They also connect computers on those networks to the Internet. Routers enable all networked computers to share a single Internet connection, which saves money.

Network Routers

The router acts a dispatcher. It analyzes data being sent across a network, chooses the best route for data to travel, and sends it on its way.

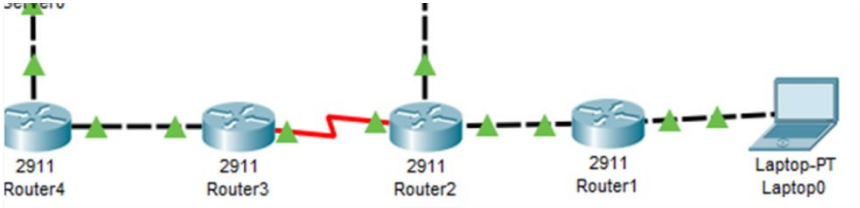
Routers connect your business to the world, protect information from security threats, and can even decide which computers receive priority over others.



Consider a router as an air traffic controller, and consider data packets as planes flying to various airports (or networks). Each packet must be directed as quickly as possible to its destination, just as each plane has a distinct destination and travels a distinct route. A router assists in guiding data packets to their intended IP address, just like an air traffic controller ensures that aircraft reach their destinations without getting lost or experiencing significant disruptions en route.

An internal routing table, which is a list of routes to different network destinations, is used by a router to effectively direct packets. In order to determine the destination of a packet, the router first scans its header. Then, it consults the routing table.

An internal **routing table**, which is a list of routes to different network destinations, is used by a router to effectively direct packets. The router first analyzes the header of a packet to establish its destination before consulting the routing table to determine the fastest route there. The packet is subsequently forwarded to the following network along the path.



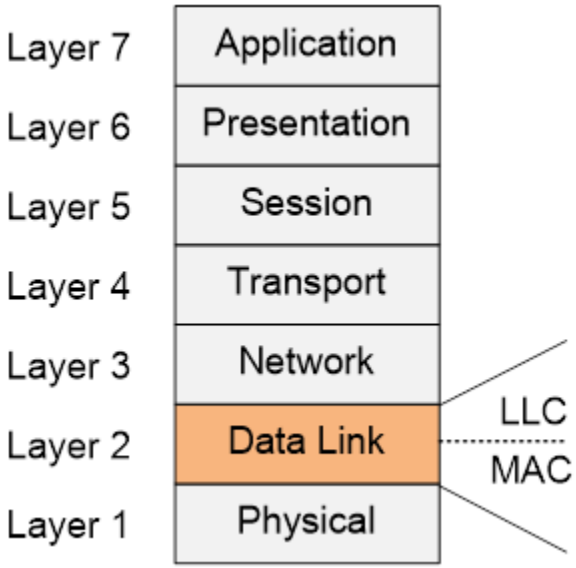

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter a
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.200.100.0/30 is directly connected, Serial0/3/0
L       10.200.100.1/32 is directly connected, Serial0/3/0
     172.16.0.0/16 is variably subnetted, 5 subnets, 2 masks
C       172.16.16.0/24 is directly connected, GigabitEthernet0/0
L       172.16.16.2/32 is directly connected, GigabitEthernet0/0
D       172.16.17.0/24 [90/2170112] via 10.200.100.2, 00:16:38, Serial0/3/0
C       172.16.18.0/24 is directly connected, GigabitEthernet0/1
L       172.16.18.1/32 is directly connected, GigabitEthernet0/1
```

Network Switches

Network switches are the foundation of most business networks. A switch acts as a controller, connecting computers, printers, and servers to a network in a building or a campus.

Network switches connect network devices to host computers and allow a large number of devices to share a limited number of ports. They increase network capacity and speed by examining and filtering data packets. Switches also regenerate forwarded packets, reducing collision rates and permitting the use of additional nodes.

Data Link layer 2 is where a network switch physically operates In the Open Systems Interconnection (OSI) architecture. It receives packets sent by devices connected to its physical ports and sends them to the targets of the packets. Newer switches can also function at Layer 3, or the Network Layer.



Network Switches

Switches allow devices on **your network** to communicate with each other, as well as with other networks, creating a network of shared resources. Through information sharing and resource allocation, switches save money and increase productivity.

There are two basic types of switches to choose from as part of your networking basics:

- on-premises
- cloud-managed

A **managed on-premises** switch lets you configure and monitor your LAN, giving you tighter control of your network traffic.

A **cloud-managed switch** can simplify your network management. You get a simple user interface, multisite full-stack management, and automatic updates delivered directly to the switch.

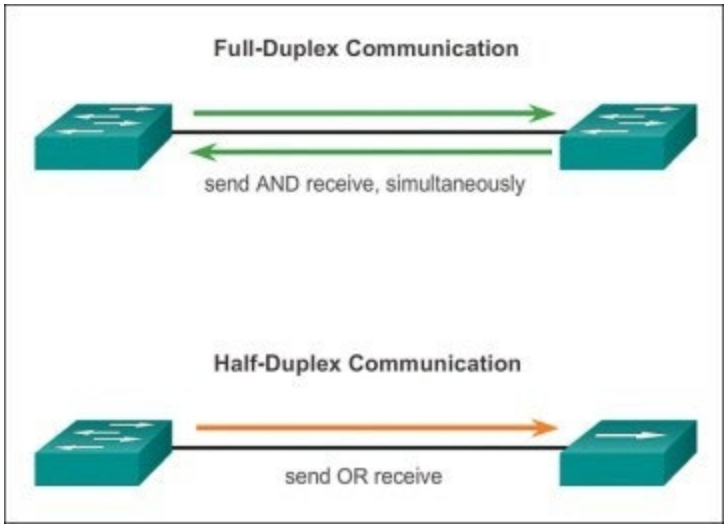
A switch records the media access control (MAC) address of a connected device, which is a code built into the network interface card of the connected device (NIC). The MAC address is used by the switch to determine which device is sending outgoing packets and where to send incoming packets.

The network layer's (Layer 3) IP address can change, but the MAC address identifies the physical device and is static. (Consider an IP address as a license plate and a MAC address as the VIN number on a car.)

The switch reads the header, then matches the destination address or addresses and sends the packet out through the appropriate ports that lead to the destination devices.

Most switches have **full-duplex** functionality, in which packets coming from and going to a device have access to the full bandwidth of the switch connection, to lessen the possibility of network traffic to and from a switch and a connected device colliding at the same time. (Consider two individuals conversing using a smartphone as opposed to a walkie-talkie.)

https://www.youtube.com/watch?v=LMRSS7ZYM50



Network hardware is crucial for interaction and communication as industry grows. These specialized hardware elements link to one another and allow a network to run smoothly and effectively enable industries to grow and expand across the globe, making sharing software, data and resources sharing more efficient.

Compromising or taking down any one of these devices can disrupt the communication process, so it is crucial to understand the functionality of these devices so we can better understand how to protect them.